

Color Display on Moving Object Trajectories Using High-Speed Projection

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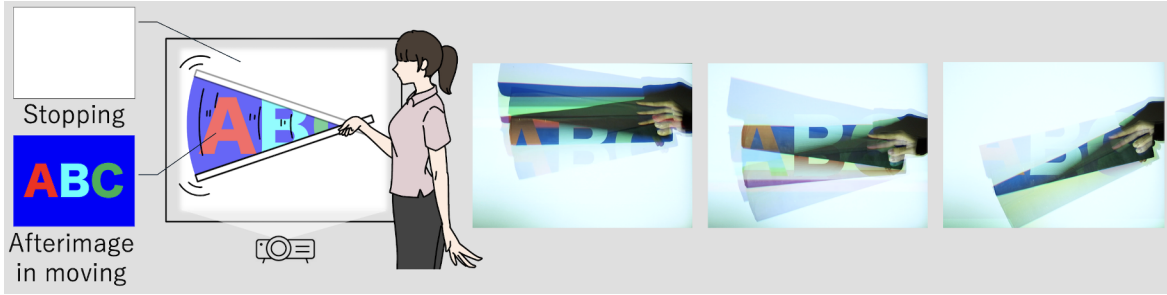


Figure 1: Our projection method controls the color along the trajectory of moving objects by leveraging the disruption of the afterimage effect. The text "ABC" is embedded in the projected white light. During motion, the text appears along the object's trajectory in the target color, while the white light appears when the object stops.

ABSTRACT

We propose a method to display target colors along the trajectories of moving objects using high-speed projection. This is achieved by leveraging the selective disruption of the afterimage effect, ensuring the target color appears prominently along motion trajectories. In our method, a high-speed projector is used to display multiple frames, consisting of target color frames and a single complementary color frame. This ensures that the target color appears prominently along the motion trajectories of the objects, where the afterimage effect is selectively disrupted, while maintaining color perception when the objects stop. This method enables novel interactions by dynamically adapting visual effects to the motion of objects.

Index Terms: high speed projection, afterimage effect, visual perception, computer vision, human-computer interaction, augmented reality

1 INTRODUCTION

Human visual perception integrates light over short periods, allowing for the recognition of colors and shapes. This ability is widely utilized in technologies such as DLP projectors and displays, where RGB light is used to create a broad range of colors. During temporal color mixing, a phenomenon known as color-breakup can occur when an object moves rapidly or during eye saccades. In such cases, the individual RGB colors are perceived separately. To utilize this perceptual property, some methods have been proposed to induce intentional afterimages by controlling motion speed and light integration. For example, some method have been proposed that arrange 1D LED lights and use the saccade effect to make them appear as 2D images [1]. Additionally, other methods have been proposed to provide different images based on specific motion vectors [3]. These methods enable delay-free perceptual changes without the need for sensors or tracking devices. Nevertheless,

in these methods, light integration depends on the motion vector during observation, which imposes constraints on the observer's motion vector. To address this, a method was proposed to relax constraints on motion direction and speed, allowing the intentional perception of mosaic patterns during walking [2]. Despite this advancement, previous methods faced a trade-off between achieving strong visual effects by precisely controlling the motion vector and relaxing those constraints at the cost of reduced color expressiveness. In this paper, we propose a high-speed projection method that enhances color control of motion trajectories of objects with minimal motion constraints. Our approach leverages the disruption of the afterimage effect to display specific colors or patterns along the trajectories of moving objects that shield the projection. Unlike previous methods, our method allows colors to appear prominently in the motion trajectories without requiring sensors, wearables, or specialized lenses. By dynamically controlling visual effects based on motion, our method facilitates diverse and engaging visual experiences in public spaces while minimizing constraints on motion direction and speed.

2 METHOD AND IMPLEMENTATION

The proposed method uses high-speed projection and leverages the disruption of the afterimage effect to display target colors along the trajectories of moving objects. The key idea is to increase the proportion of target color frames in the projection sequence, allowing the target color to appear prominently where the motion of the object disrupts the integration of light. When the object stops, complementary color frames are added to display a different image, enabling dynamic color changes based on motion. In the proposed method, projected frames are designed to create specific visual effects. The target color image I_t , visible during motion, is achieved by increasing the proportion of its frames. The image I_s is perceived when the object stops. Here, F_d represents the target color frames, and F_c represents the complementary color frame. These frames are calculated in pixels, where f_d , f_c , i_t , and i_s denote the pixel values of F_d , F_c , I_t , and I_s , respectively. To achieve the perception of these images, the projected frames are designed as follows:

$$f_d = \frac{i_t}{T-1}, \quad f_c = i_s - (T-1)f_d \quad (1)$$

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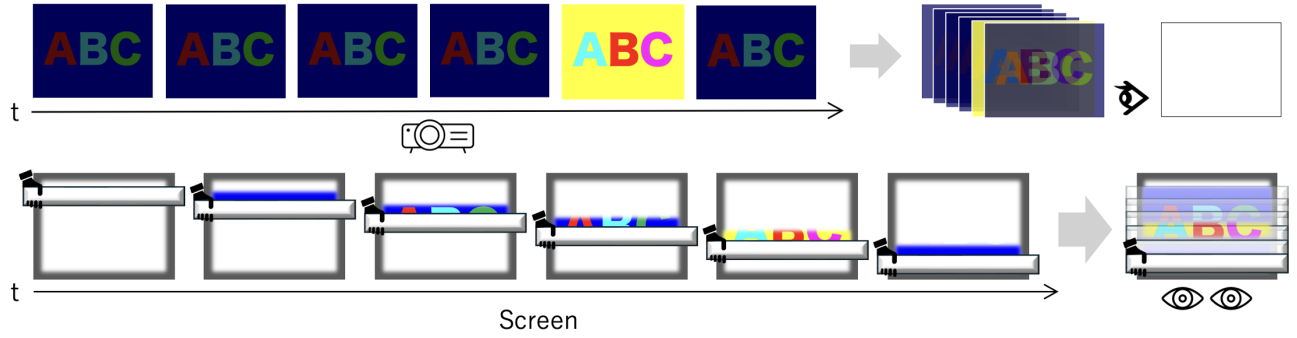


Figure 2: (Top) In the proposed method, the image is decomposed into multiple target color frames F_d and one complementary color frame F_c . These frames are superimposed in the visual perception. (Bottom) Increasing the proportion of target color in the projected frames causes the target color to appear along the trajectory of the moving object.

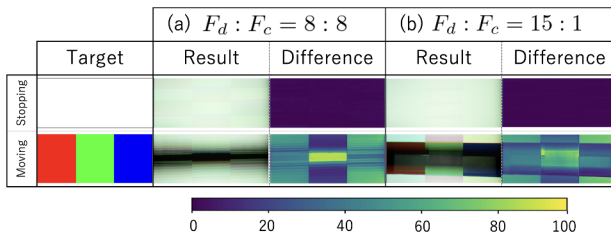


Figure 3: Evaluation results of evaluation of the effect of decomposition ratio. Colormap values are shown in the color bar at the bottom.

By repeatedly projecting the $T - 1$ frames of F_d and the one frame of F_c , all frames are superimposed due to the afterimage effect, and the image I_s is perceived when stopping (Fig. 2). The system uses a high-speed DynaFlash projector controlled by a PC with an RTX-3080 GPU. A total of 16 frames ($T = 16$) are prepared, consisting of 15 frames of F_d and one frame of F_c . These frames are repeatedly projected at 960 fps using the high-speed projector to achieve the desired visual effect.

3 EVALUATION

We evaluated the proposed method by testing the hypothesis that increasing the proportion of target color frames enhances the clarity of the target color along the trajectory of moving objects by disrupting the afterimage effect. To test this, the image I_s was divided into 16 frames with two frame ratios: (a) $F_d : F_c = 8 : 8$ and (b) $F_d : F_c = 15 : 1$. Frames were projected at 960 fps, with results captured using a SONY ZV-E10 camera at 60 fps. For stopping, the camera was fixed; for moving, a black rod fell naturally in front of the screen. The projected frames showed white light when stopping and RGB colors (left to right) when moving. Color differences between the projection targets and observed results were calculated using the CIEDE2000 metric [4] and visualized as colormaps.

Figure 3 shows the projection results and color difference maps. In the stopping condition, both ratios produced images that closely matched the target. In the moving condition, distinct target colors were observed along the trajectory of the rod, where the afterimage effect was disrupted by the motion of the rod. In condition (a), it was difficult to accurately control the trajectory color near the rod; however, in condition (b), the target color was clearer, with these differences being reduced. These results demonstrate that a higher proportion of target color frames enhances the ability to selectively display the target color along the trajectory of moving objects.

4 DISCUSSION

Speed of Moving Objects For the method to work effectively, the object need to move faster than the afterimage duration of the human eye. Additionally, the afterimage disruption needs to last longer than the minimum time humans need to recognize an object. Choosing a motion speed of objects that meets both conditions is required.

Projection Frequency Temporal color mixing relies on a high projection frequency to avoid flicker. The optimal frequency depends on factors such as ambient light and visual sensitivity. In this study, a projection frequency of 960 fps and 16 frames were used, but adjustments are needed based on environment and image conditions.

5 CONCLUSION

In this study, we proposed a high-speed projection method that selectively displays target colors along the trajectories of moving objects. The proposed method rapidly projects multiple target color frames and a single complementary color frame, emphasizing the target color along the motion trajectories. Since the proposed method does not rely on any tracking devices, it is suitable for public space interactions.

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